TEMPORAL VISION

A controlled study of ordered visual observations for human action recognition

Technical Report | HMDB51 local subset | 1 November 2026

Abstract

This project investigates whether a sequence of video frames provides useful information beyond a single image for human action recognition. A 950-video, nine-class HMDB51 subset was sampled into ordered JPEG observations. Frozen CLIP and SigLIP encoders generated frame-level representations, which were analyzed and used to train single-frame, order-agnostic pooling, and temporal classifiers. Controlled tests examined observation count, sequence order, and embedding-space corruption. An inference-only vision-language model (VLM) was also evaluated on all five sequence-understanding capabilities listed in the assignment. The strongest result was a CLIP Transformer Encoder at 98.73% test macro-F1; however, the SigLIP single-frame baseline outperformed its temporal alternatives at 97.46%. Temporal modeling therefore helped in one encoder/model pairing, but additional frames did not guarantee improvement and measured gains cannot automatically be attributed to chronological motion reasoning.

Project at a glance

950 videos	9 classes	6,662 frame observations
2 frozen encoders	12 classifiers	143 test videos

Main findings

• **CLIP:** Transformer Encoder reached 0.9873 macro-F1, +3.94 percentage points over the CLIP single-frame baseline (0.9479).
• **SigLIP:** single-frame baseline reached 0.9746; best temporal model (Temporal CNN) reached 0.9621 (–1.25 points).
• **Order:** CLIP Transformer scored 0.9873 chronological, 0.9811 reversed, and 0.9805 mean across 20 shuffled permutations.
• **VLM:** 114/143 valid structured outputs (79.72%); class-label coverage 66.43%; emitted-label accuracy 85.26%; accuracy with abstentions counted wrong 56.64%.

Scope of claims

Classifier metrics are from one fixed video-level split of this partial nine-class subset. No statistical-significance claim is made. The VLM is a separate prompted inference experiment, not a directly equivalent supervised benchmark.

1. Dataset & Sequence Analysis

Dataset source and subset

The working data is a partial local download of the HMDB51 mirror identified in the notebook as *Sina272/hmdb51-v2*. HMDB51 contains 51 action categories, but the local subset has 950 AVI files across nine available categories; it is not a complete or class-balanced representation of the original dataset.

Action class	Videos
brush_hair	107
cartwheel	107
catch	102
chew	109
clap	130
climb	108
climb_stairs	112
dive	127
draw_sword	48
Total	950

Measured property	Notebook result
Local files / classes	950 / 9
Video storage	325.05 MB
Metadata rows / local matches	6,766 / 950
Opened / yielded decoded frames	950 / 950
Reported/decoded frame-count match	4 matches; 946 discrepancies
Frame rate	30 FPS
Duration	Mean 3.56 s; median 2.63 s; range 0.83–24.23 s
Dimensions	Height 240 px; width 176–592 px (mean 333.36 px)

Frame sampling and integrity

The extraction pipeline requested up to eight approximately uniformly spaced frames per source video and preserved original frame index, sampled position, and timestamp (derived from FPS). It saved 6,662 JPEGs across all 950 videos: 938 videos have seven saved frames and 12 have eight. The manifest is sorted by video ID and original frame index. An explicit ordering check over 50 inspected videos found zero index-order failures and zero timestamp-order failures. Notebook 02 independently confirmed 950 ordered sequences with no duplicate frame indices.

Decoder-count qualification

All 950 AVI files opened and yielded decoded frames, and all 950 were processed by frame extraction. However, OpenCV's reported frame count matched the number decoded for only four files; 946 had a discrepancy. This is a reported-versus-decoded count inconsistency, not 946 failed videos. The report does not treat the reported frame count as exact ground truth.

Temporal observations

Frame strips and early/middle/late comparisons were generated. The notebook proposes cartwheel, climb, climb_stairs, and dive as actions whose progression may matter; catch and clap as potentially ambiguous in a still image; and some brush_hair, chew, or draw_sword examples as potentially recognizable from appearance or object context. These are hypotheses, not completed quantitative annotations: the saved sequence-inspection worksheet is empty. Therefore, no specific clip is presented here as a manually verified example. A sparse strip may also miss brief events between sampled frames.

2. Visual Representations & Architecture

Frozen frame encoders

Each saved JPEG is independently passed through a pretrained image encoder to produce one vector per frame. The embedding stores retain a row-aligned manifest with video ID, label, original frame index, sampled position, timestamp, and image path. Both encoders produce L2-normalized vectors. Cosine similarity is therefore equivalent to the dot product up to floating-point precision. CLIP and SigLIP are analyzed independently; their raw vectors are not directly comparable because dimensions differ.

Encoder	Model identifier	Embedding matrix	Dimension
CLIP	openai/clip-vit-base-patch32	6,662 × 512	512
SigLIP	google/siglip-base-patch16-224	6,662 × 768	768

Embedding analysis

The notebook computes adjacent-frame cosine similarity, pairwise similarity between mean-pooled video vectors, PCA projections of chronological per-video embedding trajectories, and mean/max temporal pooling. There are 5,712 adjacent pairs per encoder. Mean adjacent similarity is 0.9257 for CLIP (median 0.9368; standard deviation 0.0502) and 0.9341 for SigLIP (median 0.9453; standard deviation 0.0499). High similarity indicates similar encoded visual content in adjacent samples; lower similarity may reflect pose, object, viewpoint, lighting, or scene changes. It is not a direct measurement of physical motion.

Mean/max pooled-vector cosine similarity averages 0.9042 for CLIP and 0.9202 for SigLIP. This is descriptive only; it does not establish which pooling strategy is better. Mean pooling discards chronological order and serves as an order-agnostic control.

Split and training protocol

A shared stratified split by video ID is reused across both encoders and all models: 664 train, 143 validation, and 143 test videos (70% / 15% / 15%), with seed 42. Frames from one video never cross splits. Variable-length sequences are padded only within batches, with true lengths retained.

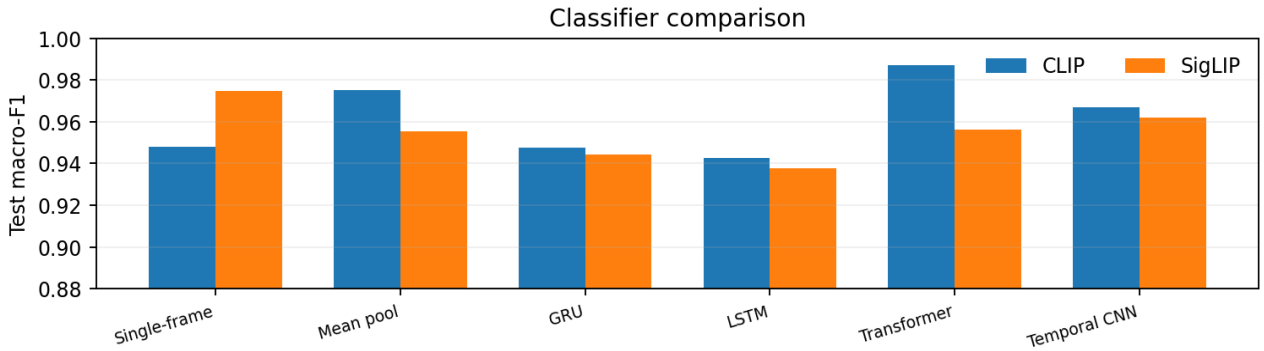
All classifiers use frozen precomputed embeddings; only the compact classification modules are trained. The single-frame baseline selects the deterministic middle frame. Mean pooling averages all frame vectors. Training uses weighted cross-entropy, AdamW (learning rate 1e–3; weight decay 1e–4), batch size 32, dropout 0.25, up to 60 epochs, and early stopping (patience 10) on validation macro-F1. Gradient norms are clipped at 1.0; ReduceLROnPlateau adjusts the learning rate.

Architectures

Single-frame and mean-pooling approaches use an MLP (input → 256 → 128 → 9, ReLU/dropout). GRU/LSTM project each frame to 128 dimensions and use one unidirectional recurrent layer (hidden size 128); the final valid hidden state is classified. The Transformer projects to 128, adds learned positional embeddings, applies a two-layer encoder (four heads; feed-forward dimension 256), masks padding, and classifies the masked mean of valid outputs. Temporal CNN projects to 128 channels, applies two residual 1D convolution blocks (kernel 3; dilations 1 and 2), then masked temporal averaging and a classifier.

3. Single-Frame vs Temporal Model Results

Twelve models were trained: six approaches for each encoder (single-frame MLP, mean-pooling MLP, GRU, LSTM, Transformer Encoder, Temporal CNN). Checkpoints were selected by validation macro-F1 and evaluated on the held-out test split. Macro-F1 is the primary metric because the nine classes are unevenly represented.



Approach	CLIP accuracy	CLIP macro-F1	SigLIP accuracy	SigLIP macro-F1
Single-frame	94.41%	0.9479	97.20%	0.9746
Mean pooling	97.20%	0.9753	95.10%	0.9555
GRU	94.41%	0.9478	93.71%	0.9443
LSTM	93.71%	0.9425	93.71%	0.9379
Transformer Encoder	98.60%	0.9873	95.10%	0.9561
Temporal CNN	96.50%	0.9668	95.80%	0.9621

Primary result

For CLIP, the Transformer Encoder achieved 98.60% accuracy and 0.9873 macro-F1. Relative to the CLIP single-frame baseline (94.41%; 0.9479), this is +4.20 accuracy points and +3.94 macro-F1 points. It also exceeded CLIP mean pooling (0.9753) by 1.20 macro-F1 points.

For SigLIP, the single-frame baseline was strongest (97.20% accuracy; 0.9746 macro-F1). The best temporal model was Temporal CNN (95.80%; 0.9621), 1.25 macro-F1 points below the single-frame baseline. SigLIP Transformer reached 0.9561.

Interpretation

The results do not support a universal claim that temporal models outperform static models. CLIP paired effectively with the Transformer in this split, whereas SigLIP already provided a strong single-frame representation and temporal models did not improve it. CLIP mean pooling was also strong, suggesting that aggregating multiple observations can help without necessarily using their order. Encoder and classifier must be considered jointly.

The test set contains 143 videos. These are point estimates from one split; no confidence intervals, repeated-seed estimates, or significance tests are reported. Near-ceiling scores on this small subset require caution.

4. Temporal Evaluation

Three controlled evaluation families use the same 143 test videos and saved checkpoints. The evaluation notebook reuses the exact split and does not retrain models or regenerate embeddings.

A. Progressive observation

For each video, k = 1…7 embeddings are selected approximately uniformly across the timeline. k=1 selects the middle frame. Temporal checkpoints are evaluated on reduced sequences without retraining and compared with a fixed single-frame reference. All 143 videos are included at every budget. Results are non-monotonic: more observations do not consistently increase macro-F1. For example, CLIP GRU scores 0.9007 at one observation, 0.9497 at two, 0.9470 at three, 0.9749 at four, and 0.9477 at five.

B. Temporal order

The same trained temporal models are evaluated chronologically, in reverse, and under 20 independent random permutations per video. CLIP Transformer macro-F1 is 0.9873 chronological, 0.9811 reversed, and 0.9805 averaged across 20 shuffled permutations (standard deviation 0.0052). The chronological advantage over the shuffled mean is 0.68 points. SigLIP Transformer scores 0.9561 chronological, 0.9496 reversed, and 0.9534 shuffled (standard deviation 0.0034). Across architectures, altered order does not consistently reduce performance; some shuffled/reversed scores are slightly higher. This indicates modest or model-dependent order sensitivity, not proof that order is irrelevant.

C. Observation corruption

Because this stage uses cached embeddings rather than source pixels, it does not apply image blur or pixel-space occlusion. It tests: frame dropping (randomly retaining 75%, 50%, or 25% of frames, preserving order and at least one frame); feature masking (zeroing 25%, 50%, or 75% of embedding dimensions); and Gaussian noise (standard deviations 0.1, 0.25, and 0.5). The same trained weights and held-out videos are used.

Encoder/model	Clean F1	Worst frame-drop	Worst feature-mask	Worst noise
CLIP single-frame	0.9479	0.9479	0.8700	0.9366
CLIP Transformer	0.9873	0.9301	0.7607	0.9697
SigLIP single-frame	0.9746	0.9746	0.8451	0.9301
SigLIP Transformer	0.9561	0.9246	0.7785	0.9561

The table reports the minimum macro-F1 across tested severities for each corruption family, as summarized in the notebook. Feature masking causes the largest reported drops in these comparisons. CLIP Transformer remains strongest on clean data but falls to 0.7607 under its worst feature-mask condition. Negative degradation values can occur on this small test set; they do not mean corruption generally improves a model.

Overall, extra observations and temporal architectures help in some conditions, but effects are not monotonic or universal. Exact order dependence is modest for the best CLIP Transformer and inconsistent across models.

5. Multimodal / VLM Evaluation

An inference-only VLM experiment covers all five requested capabilities: sequence-level description, temporal change analysis, semantic action labeling, identification of ambiguous observations, and early-versus-later comparison. Five chronological source images are sampled per test video. The notebook uses Apple Silicon MLX and performs no fine-tuning. A five-video pilot produced four valid structured outputs (80%); then all 143 held-out test videos were processed with incremental checkpointing.

VLM measure	Observed result
Evaluated test videos	143
Valid structured outputs	114 / 143 (79.72%)
Parse errors / runtime errors	29 (20.28%) / 0
Class-label coverage	66.43%
Accuracy among emitted class labels	85.26%
Accuracy with uncertain/other counted wrong	56.64%
Uncertain/other rate	33.57%
Median inference latency (metrics table)	13.50 s
Median successful-output latency (provenance)	13.20 s

Each of sequence description, temporal change analysis, semantic labeling, and early/late comparison had non-empty outputs for 79.72% of videos; ambiguity identification had 76.22%. Valid structured outputs can still abstain with uncertain/other, explaining why class-label coverage is lower. The VLM adds natural-language context and explicit uncertainty, but is slower and less reliable as a closed-set classifier in this experiment.

Classifier comparison and model-identity note

In the joined 143-video comparison, the VLM emitted class labels for 66.43% and achieved 56.64% accuracy when abstentions count as wrong. The visual classifiers report 94.41% CLIP single-frame, 98.60% CLIP Transformer, 97.20% SigLIP single-frame, and 95.10% SigLIP Transformer accuracy. These are not like-for-like benchmarks: the classifiers were supervised on the nine-class training split, whereas the VLM was prompted at inference time and can abstain. The notebook has a model identity inconsistency: prose names Qwen2.5-VL-3B-Instruct, while the executed model ID/provenance records *mlx-community/Qwen2-VL-2B-Instruct-4bit* and the provenance family field says Qwen2.5-VL-3B-Instruct. Verify this against the runtime before external submission.

6. Limitations

- Partial nine-class subset; uneven class counts; one 143-video test split. Results do not generalize to all 51 HMDB51 classes.
- Reported/decoded frame counts disagree for 946 videos, although all files yielded decoded frames.
- Only 7–8 frames are retained; sampling may miss brief events and does not represent continuous motion.
- The human sequence-inspection worksheet is empty; action-level temporal examples remain hypotheses, not recorded annotations.
- CLIP/SigLIP are frozen image encoders; temporal models operate on embeddings, not raw video or optical flow.
- One split, small test set, possible scene/background bias, and no confidence intervals or repeated seeds limit generalization.
- Corruptions are embedding-space perturbations, not pixel blur or true occlusion.
- VLM structured output has 20.28% parse errors, abstentions, possible hallucinations, and a model metadata discrepancy.

7. Future Improvements

Expand the dataset and repeat video-level splits across seeds; report paired bootstrap confidence intervals. Complete human-reviewed early/middle/late annotations. Increase or adapt temporal sampling and add pixel-space blur/occlusion tests. Investigate motion-specific features and temporal augmentation. Evaluate on another dataset or cross-environment split. Resolve the VLM model ID, strengthen JSON schema validation/retries, and manually review outputs against contact sheets. Use the VLM selectively for explanation and ambiguity rather than routine classification.

Conclusion

The project implements the full requested study: ordered dataset inspection, pretrained frame embeddings, single-frame and temporal classifiers, controlled temporal evaluation, and all five VLM capabilities. Temporal context improved the CLIP Transformer result but did not improve the SigLIP baseline; order perturbations produced modest and inconsistent changes. The evidence supports a conditional benefit from multi-observation modeling—not a blanket claim that more frames or temporal architectures are always better.

Reproducibility artifacts are saved under *data/processed/*, *embeddings/*, *Single-multiframe/*, *Temporal-Evaluation/*, and *VLM-Evaluation/*, including manifests, splits, checkpoints, predictions, confusion matrices, provenance, raw VLM responses, and analysis outputs.